Library Management System

A Python + MongoDB Desktop Application

Python | Tkinter | MongoDB | Pandas | Matplotlib

Project Overview

What is it?

A desktop GUI application for managing library books — add, view, issue, return, and analyze the book collection with ease.

Who uses it?

Librarians and administrators who need a simple, no-web, local tool to track books and their availability in real time.

Why build it?

To demonstrate how Python's Tkinter, MongoDB, Pandas, NumPy, and Matplotlib can be integrated into a real-world mini project.

Core Features

Add Book

Add new books with name & author to MongoDB

View Books

Browse all books with their availability status

Issue Book

Mark a book as issued (unavailable) to a borrower

Return Book

Mark a returned book as available again

Show Stats

View totals, available vs issued using Pandas & NumPy

Show Graph

Bar charts of book status & most issued books

System Architecture

Presentation Layer

Tkinter GUI

- Buttons & Entry fields
- Output text box
- Status label
- Main window (root)

Logic Layer

Python Functions

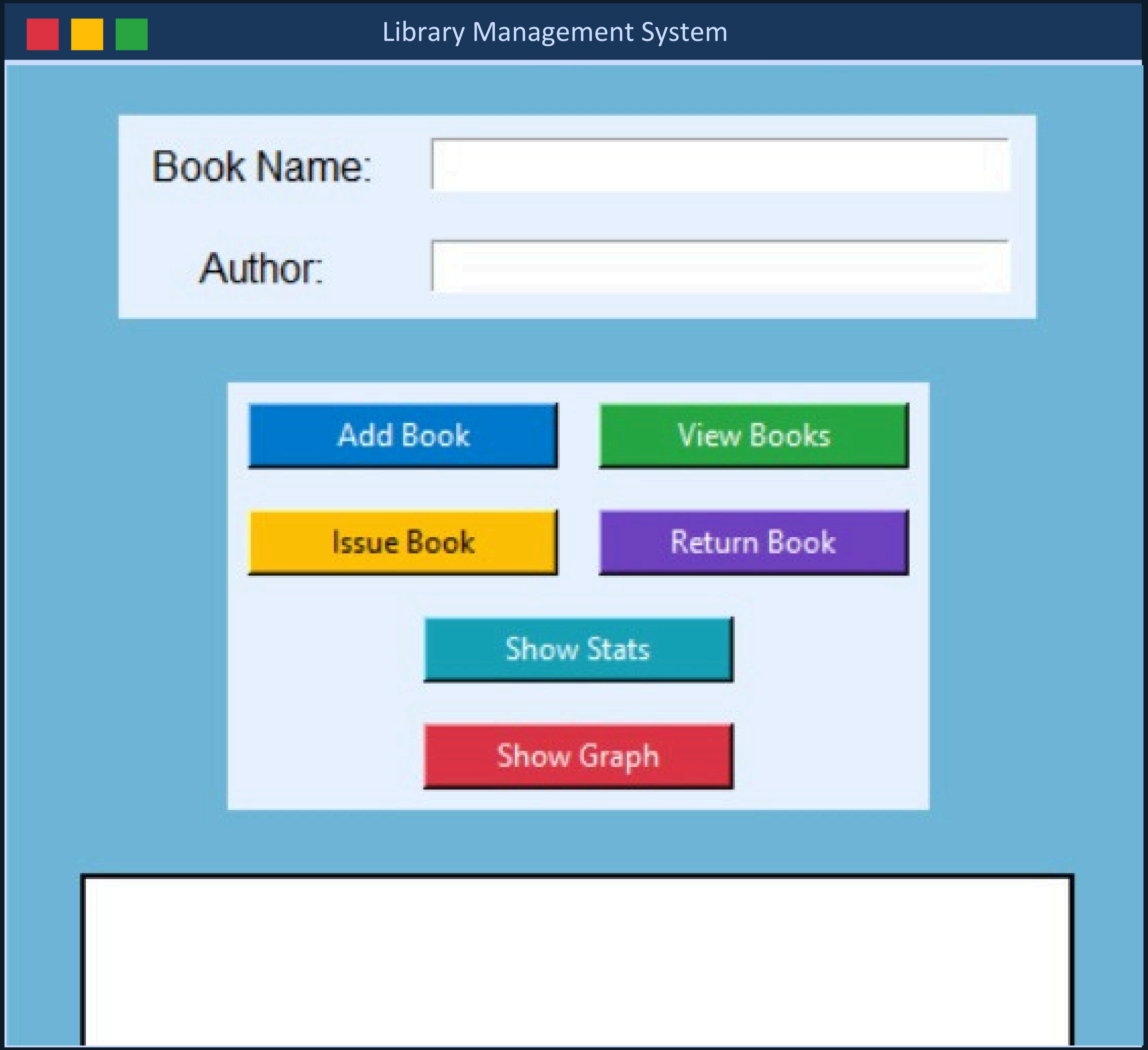
- add_book()
- view_books()
- issue_book()
- return_book()
- show_stats()
- show_graph()

Data Layer

MongoDB + Pandas

- MongoDB (books collection)
- PyMongo client
- Pandas DataFrame
- NumPy arrays

GUI Layout & Design



1. Title Header
2. Input Fields
3. Action Buttons
4. Stats & Graph
5. Output Box

Database Design (MongoDB)

Document Schema

```
"name"      :"The Alchemist",
"author"    :"Paulo Coelho",
```

Add

Issue

return

String field

Boolean field

Database

library

Collection

books

Port

27017

CRUD Operations

CREATE

`insert_one()`

→ *add_book()*

Inserts a new book document into the books collection.

READ

`find() / find_one()`

→ *view_books(), issue_book()*

Fetches all or a specific available book from MongoDB.

UPDATE

`update_one()`

→ *issue_book(), return_book()*

Toggles the 'available' field to true or false.

DELETE

`— not implemented —`

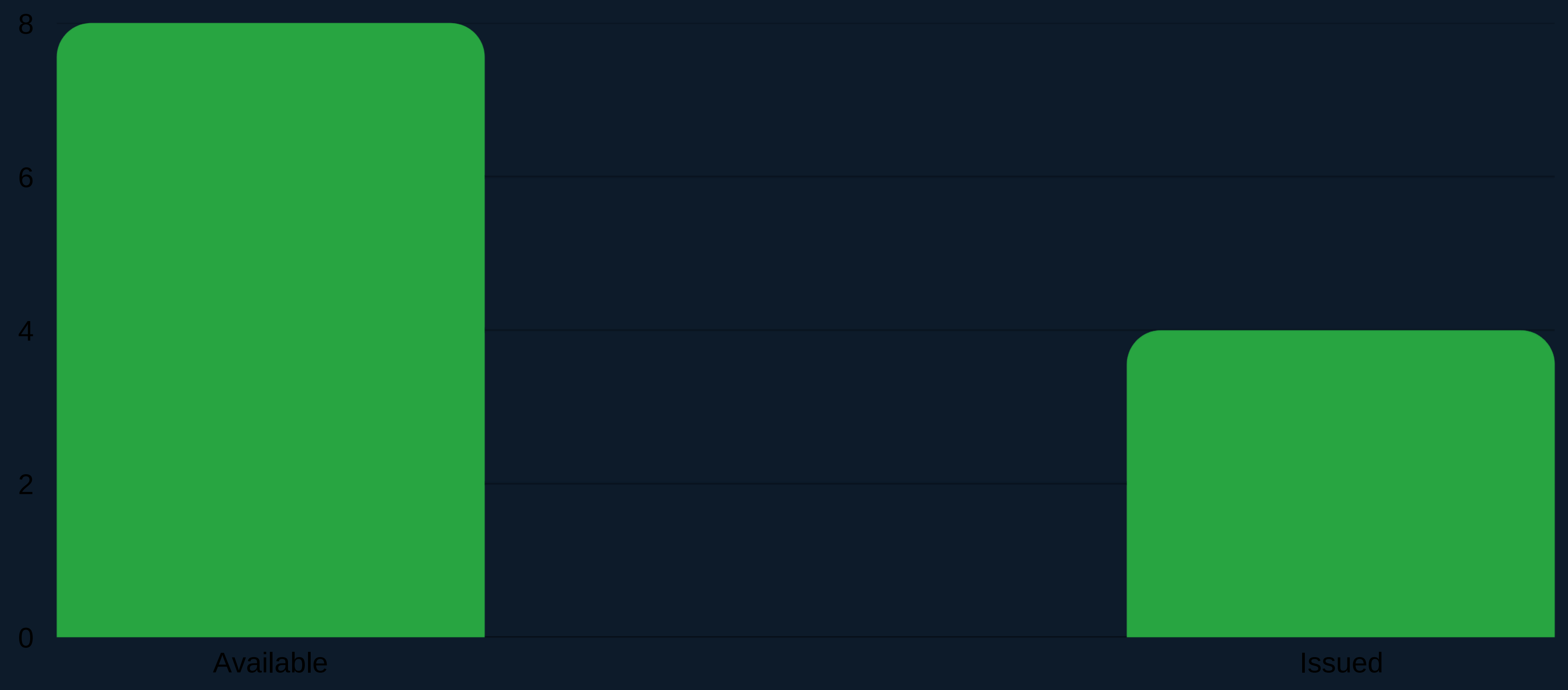
→ *future scope*

Book removal is not part of this version of the app.

Data Analysis & Visualization

Pandas+NumPy (show_stats)

```
df = pd.DataFrame(data)
total  = len(df)
avail  = len(df[df["available"]==True])
arr    = np.array([avail, issued])
```



Matplotlib (show_graph)

```
labels = ["Available","Issued"]
plt.bar(labels, values)
plt.title("Library Book Status")
plt.show()
```

Two Charts Generated:

① Book Status Chart — Available vs Issued (all books)

② Most Issued Books — Which titles were borrowed most

Both rendered as interactive Matplotlib windows.

Summary & Takeaways

- Fully functional CRUD operations via PyMongo for real-time book tracking
- Clean Tkinter GUI with 6 action buttons and a live output text area
- Data analysis with Pandas DataFrames and NumPy arrays for quick stats
- Matplotlib bar charts for visual insight into book availability & trends
- Demonstrates real-world Python library integration in a single-file project

Thank You!